SAI SRAVANA KUMARA VIGNESH ALLAM

Bremen, Germany | allam.vignesh@gmail.com | linkedin.com/in/vignesh-allam | github.com/allamvignesh

M.Sc. Artificial Intelligence and Intelligent Systems student at the University of Bremen, focused on robotics software and cognitionenabled robotics. Hands-on experience with ROS 2, custom navigation, dual-arm manipulation, Giskard/Coraplex, perceptionsystem integration, simulation, and robotics data analysis. Seeking a robotics software internship in planning, manipulation, or autonomous systems.

TECHNICAL SKILLS

Robotics: ROS 2 Jazzy, rclpy, TF2, URDF, RViz2, actions/services, Giskard, Coraplex, SLAM Toolbox, occupancy grids, A*
Programming: Python, Linux/Bash, object-oriented programming, multithreading, Git
ML & Data: OpenCV, PyTorch, Transformers, scikit-learn, NumPy, pandas, time-series analysis, ROS bag processing
Simulation & Tools: PyBullet, MuJoCo, Docker, GitHub, Jupyter, CMake, Linux

EDUCATION

M.Sc. Artificial Intelligence and Intelligent Systems — **2025 - Present**
University of Bremen, Germany | Focus: Cognition-Enabled Robotics
- Relevant coursework: ROS 2 Programming, Integrated Intelligent Systems, Advanced Machine Learning.

B.E. Computer Science and Engineering (Artificial Intelligence & Machine Learning) — **2021 - 2025**
Sathyabama Institute of Science and Technology, Chennai, India | CGPA: 8.5/10

SELECTED ROBOTICS PROJECTS

Dual-Arm Robot Manipulation - Tracy — **2026 - Present**
ROS 2, Giskard, Coraplex, UR10e, Robotiq, Python
- Integrated ROS 2 action-based control for Robotiq parallel grippers on a dual-UR10e robot using control_msgs/ParallelGripperCommand.
- Extended Coraplex real-execution support with alternative motion mappings and Giskard-compatible motion charts for hardware gripper commands.
- Developed and tested manipulation workflows for arm parking, pick-and-place, cube stacking, and integration of object poses supplied by the perception pipeline.

Tortugabot Autonomous Navigation Planner — **2026**
ROS 2, Python, SLAM Toolbox, RViz2, occupancy grids
- Built a custom A* planner without Nav2, including map-to-grid conversion, obstacle inflation, path smoothing, and turn-based waypoint extraction.
- Designed a supervisor architecture that requests path reasoning, tracks waypoint completion, and sends one waypoint at a time to a topic-based motion controller.
- Integrated live SLAM maps and RViz goal poses; designed interfaces for replanning around dynamic obstacles and following a perceived target.

Navigate-to-Grasp Mobile Manipulator — **2025 - 2026**
PyBullet, Python, LiDAR, inverse kinematics, system integration
- Contributed to an end-to-end simulated mobile-manipulation system that detects a red cylinder, navigates to the table, and executes a grasp.
- Integrated the perception module's object-pose output with navigation and manipulation components, helping coordinate data flow between the modules.
- Focused on waypoint navigation, LiDAR-based obstacle avoidance, base positioning, inverse kinematics, and phased grasp execution.

Anomaly Detection in ROS Bag Data — **2026**
ROS 2, Python, time-series analysis
- Analyzed recorded ROS topic data to identify abnormal patterns in robot and sensor behaviour through extraction, preprocessing, visualization, and anomaly scoring.

ADDITIONAL TECHNICAL PROJECT

2D Floorplan to Interactive 3D/AR Model **GitHub** — **2024 - 2025**
Django, OpenCV, Blender, GLTF, AR
- Built a Django application that converts floorplan images into interactive 3D models using OpenCV vectorization and a headless Blender pipeline.
- Documented the image-processing and 3D-generation approach in a technical research paper.

AWARDS & LANGUAGES

- 1st place, UBERTECH 2022 Capture the Flag; 2nd place, Code Combat 2023; 3rd place, OWASP 2022 Capture the Flag.
- IELTS Academic: 7.0 | English: C1 | German: A2 (basic working proficiency).